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Laboratory Report – 5

Objective

The objective of this laboratory exercise is to familiarize with QGIS (an open-source software) and LAStools for LiDAR data visualization and Digital Surface Model (DSM) generation and visualization.

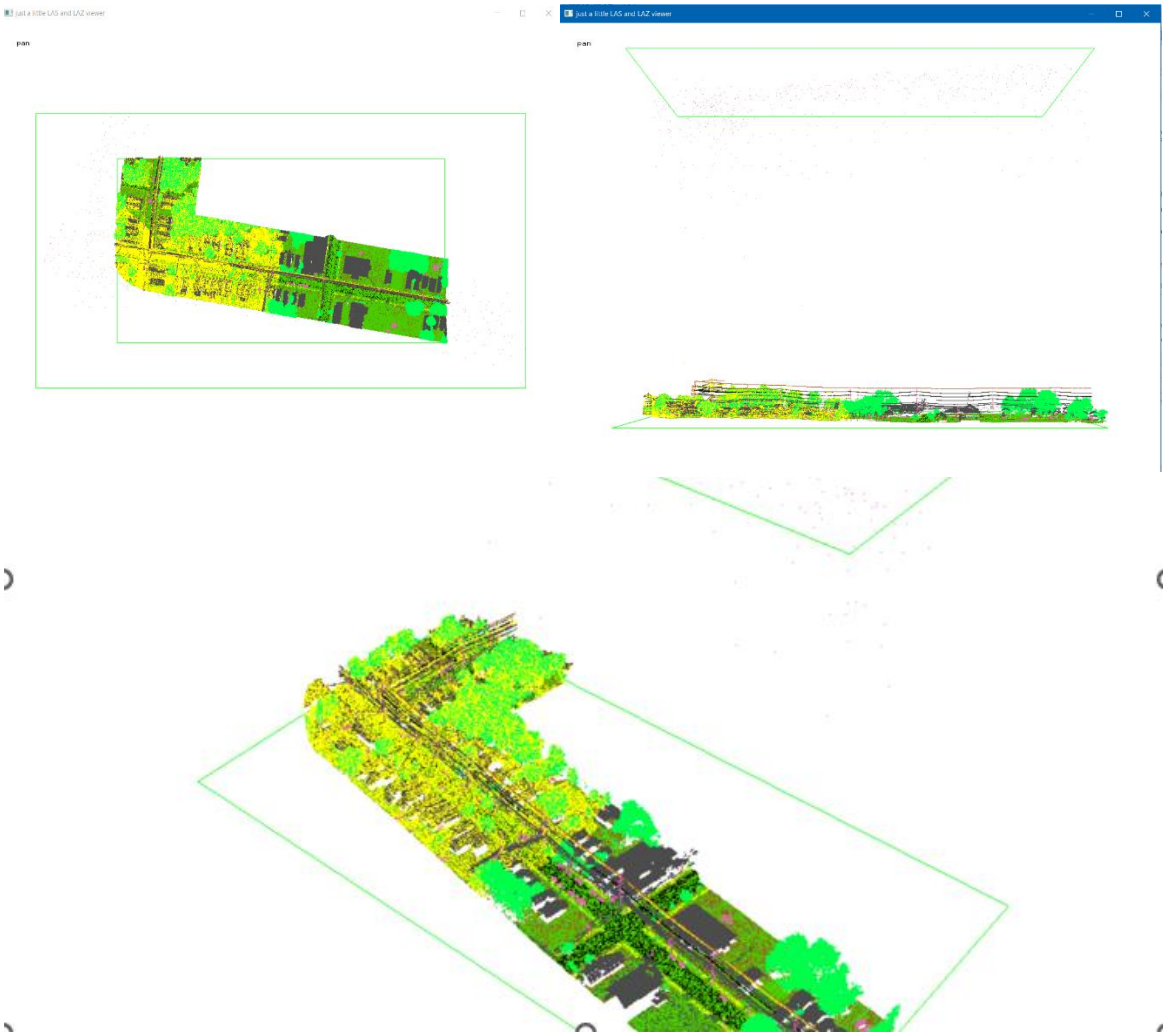
Methodology with Q-GIS and CloudCompare

• Open QGIS and from there go to processing tab. In toolbox expand LAStools and search for lasview under file-checking quality. Then we input file 1198500_276000.laz path which we want to load.

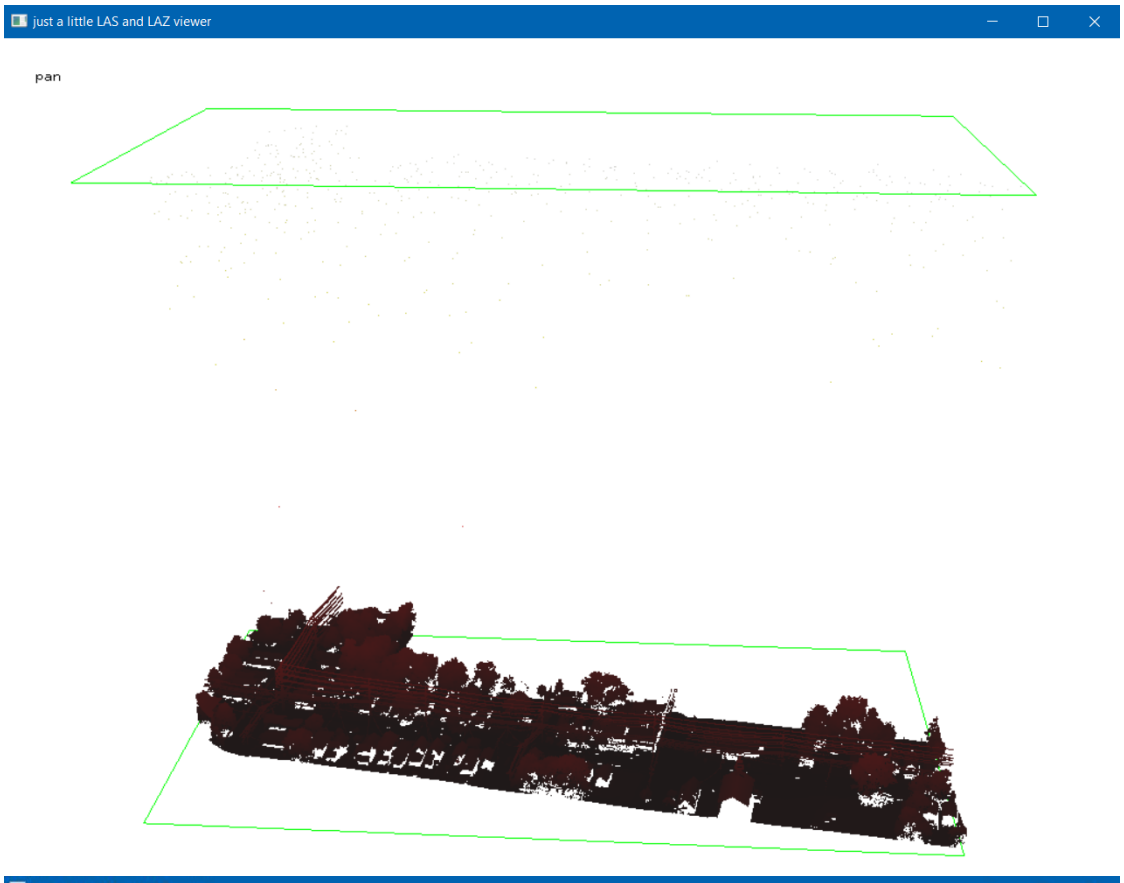
After view is generated, we try to explore various options by right clicking on the generated view window.

(i) Colour by

(i) classification

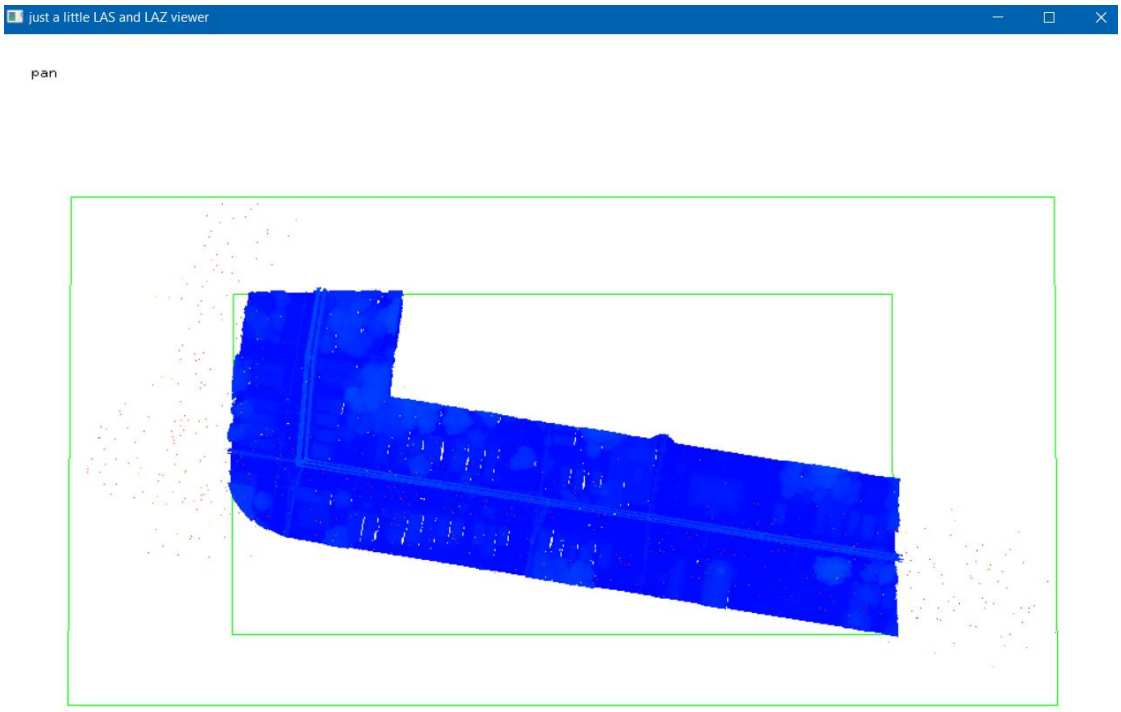


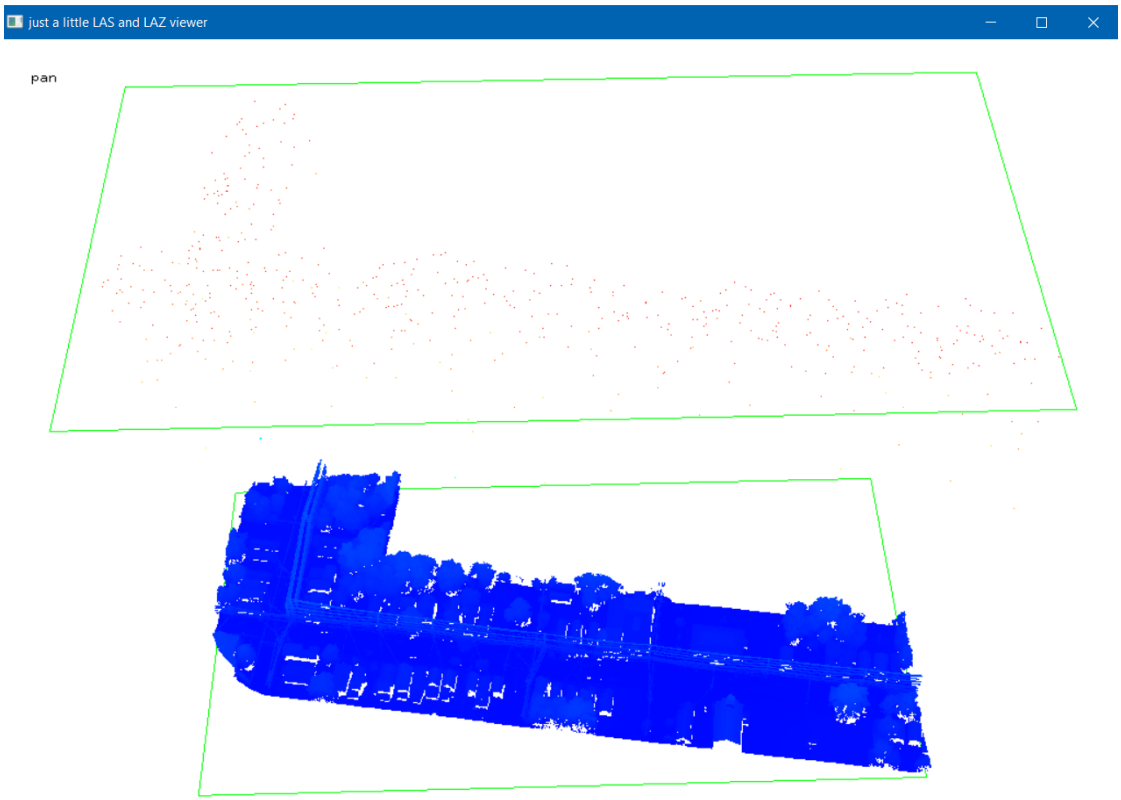
(ii) elevation (black-red-yellow-white)



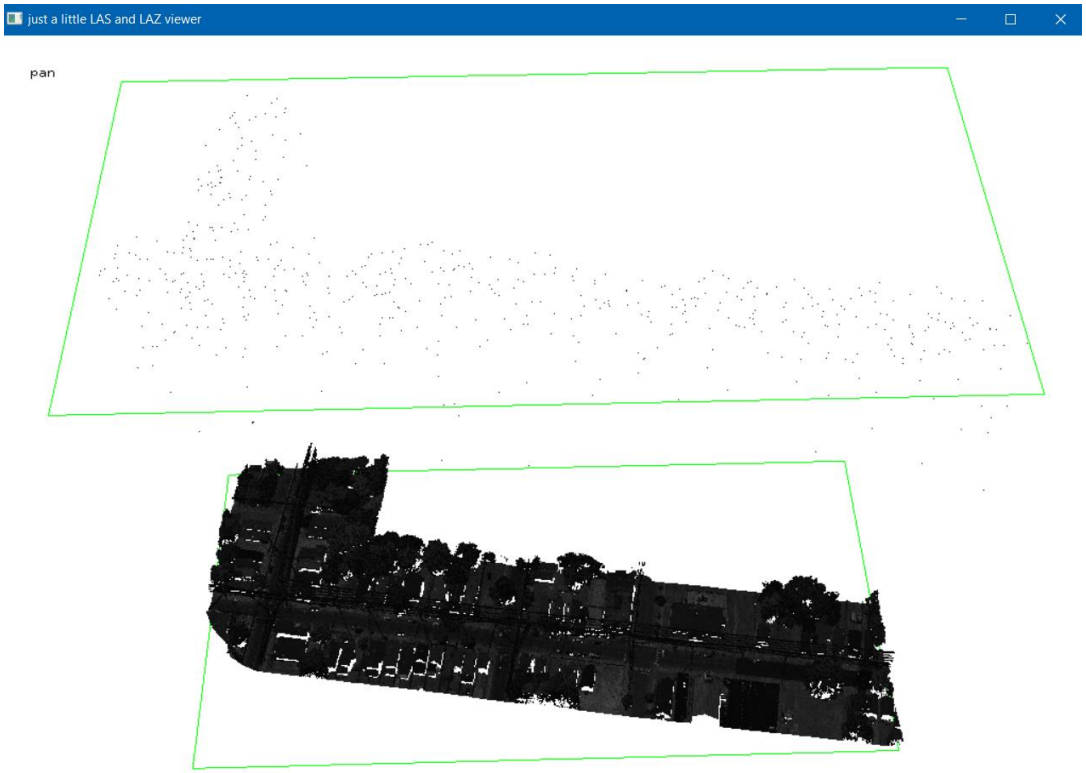


elevation (blue- yellow-red)

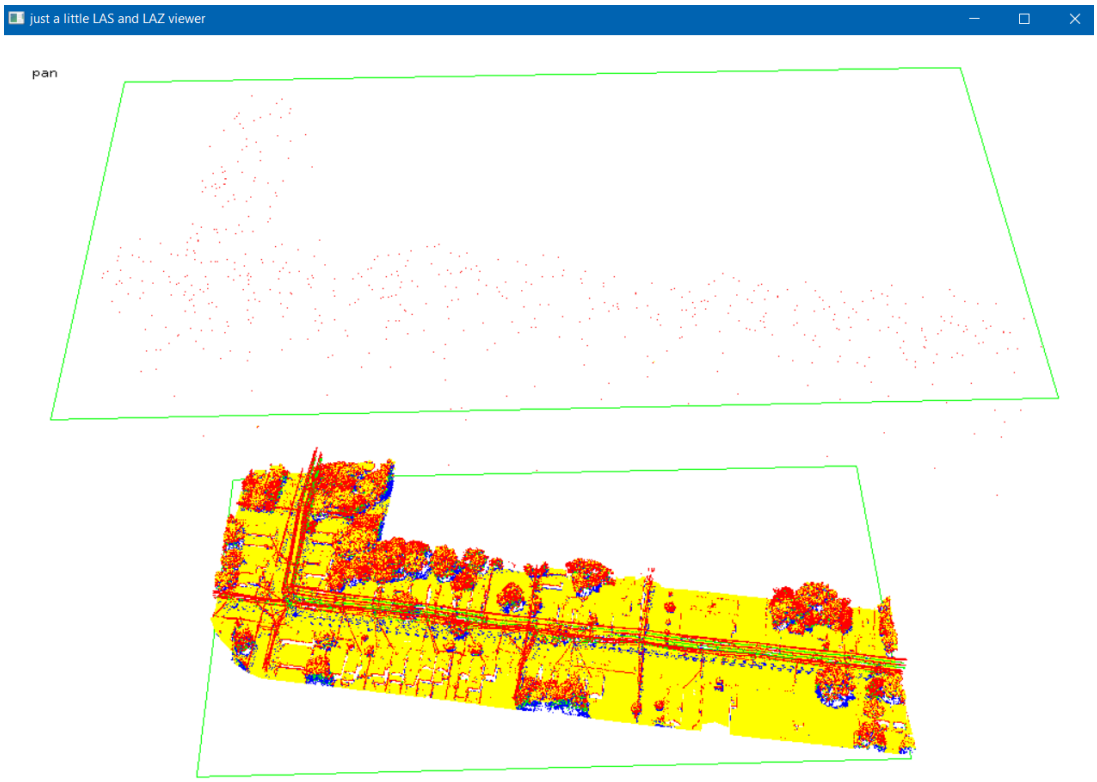




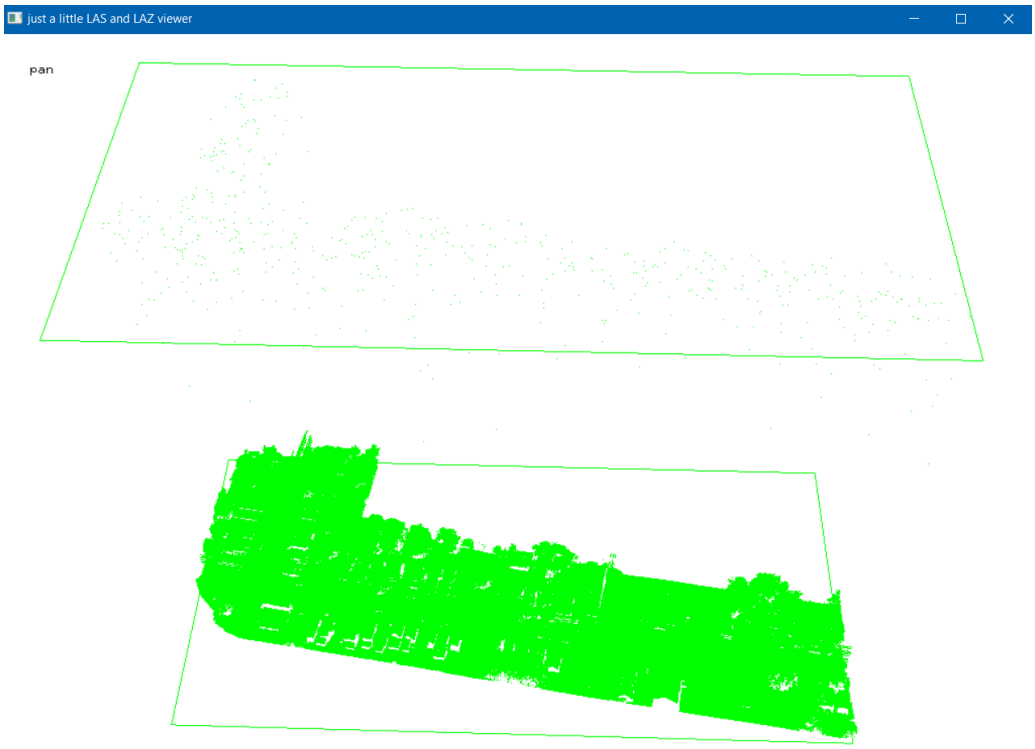
(iii) intensity


just a little LAS and LAZ viewer
pan

(iv) return

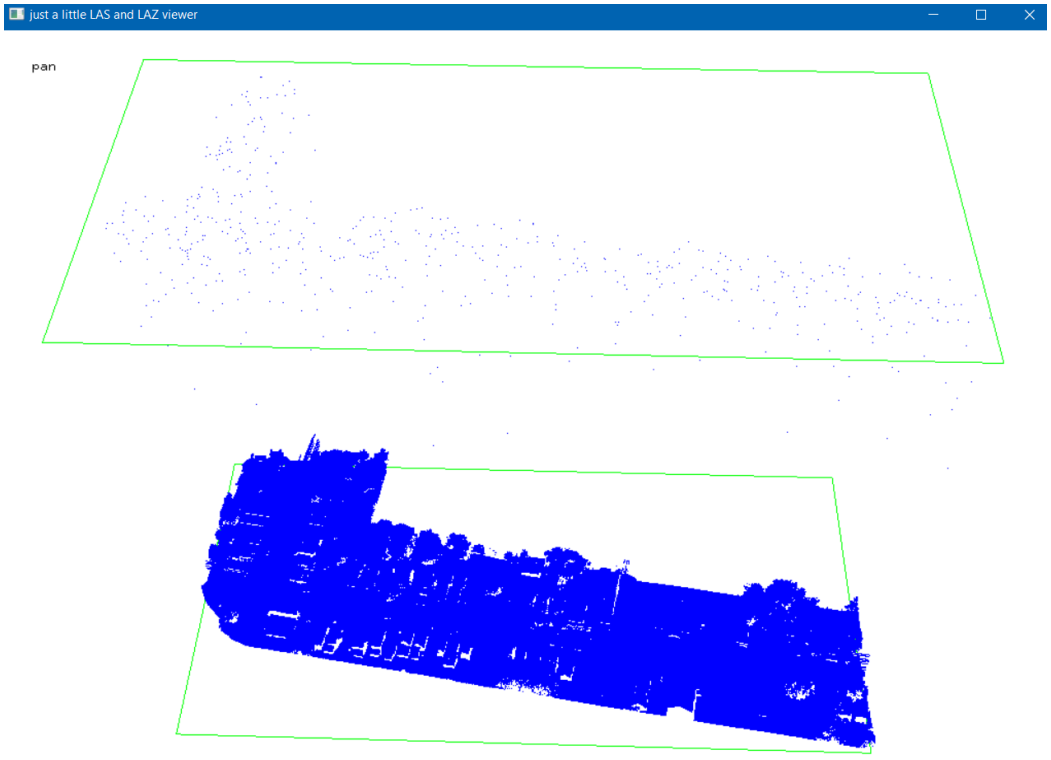

just a little LAS and LAZ viewer
pan

(ii) render only

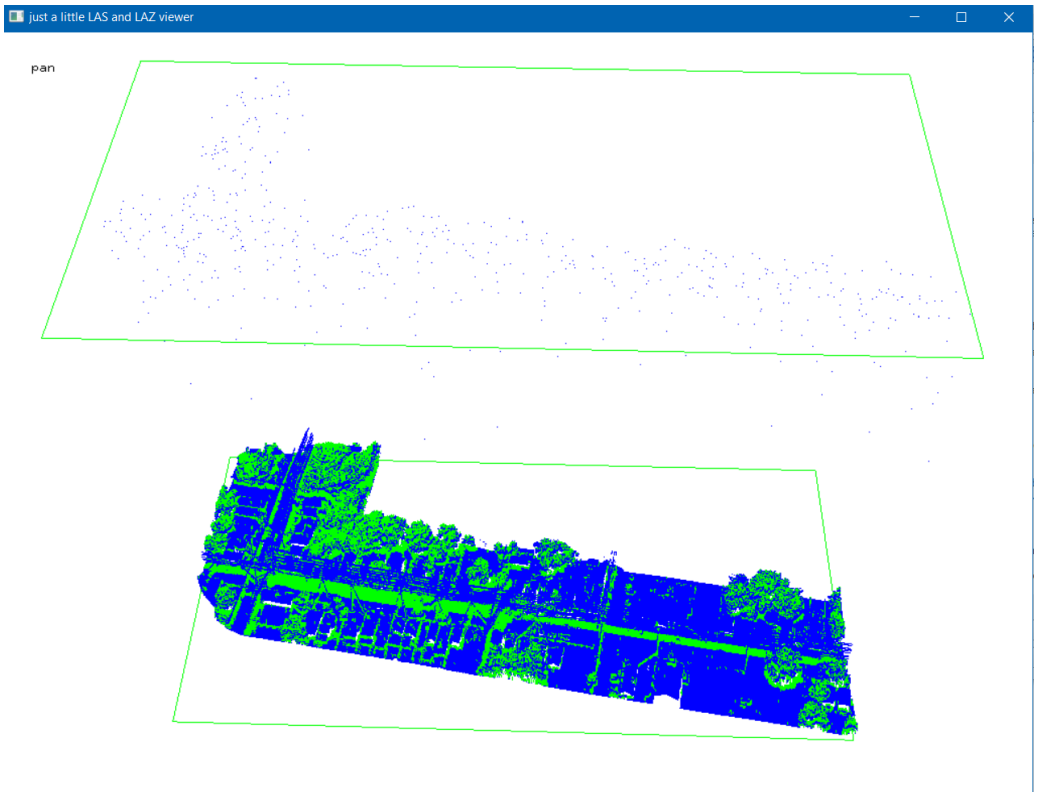


(iii) Pulse

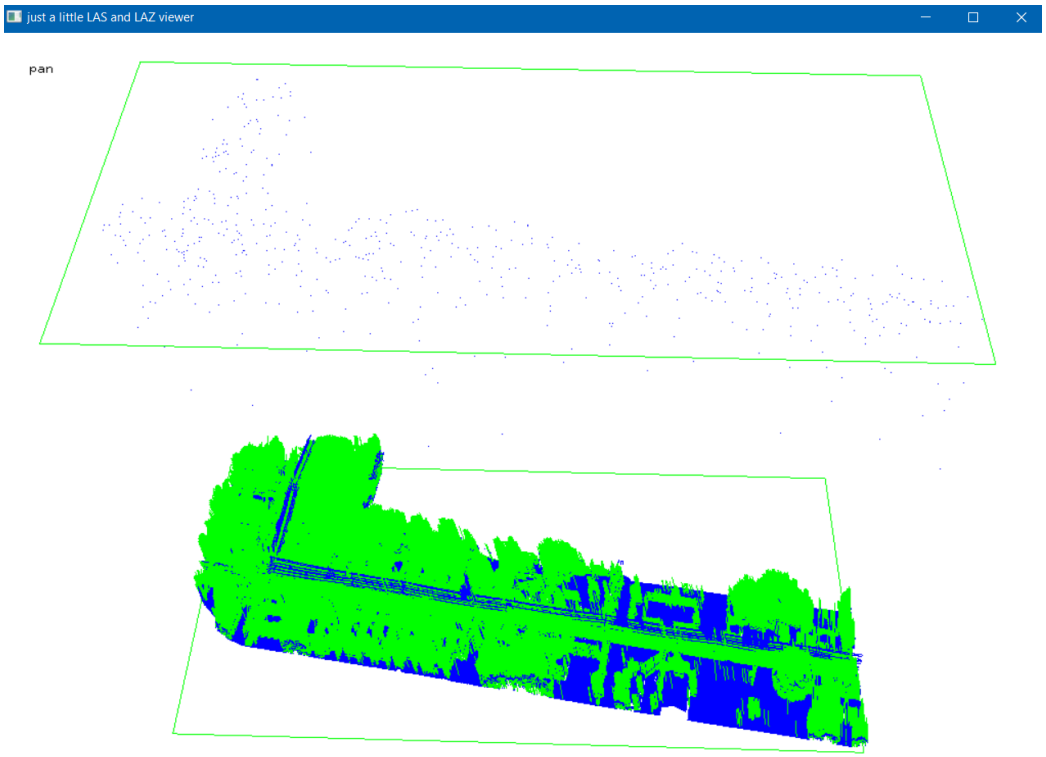
(i) 0



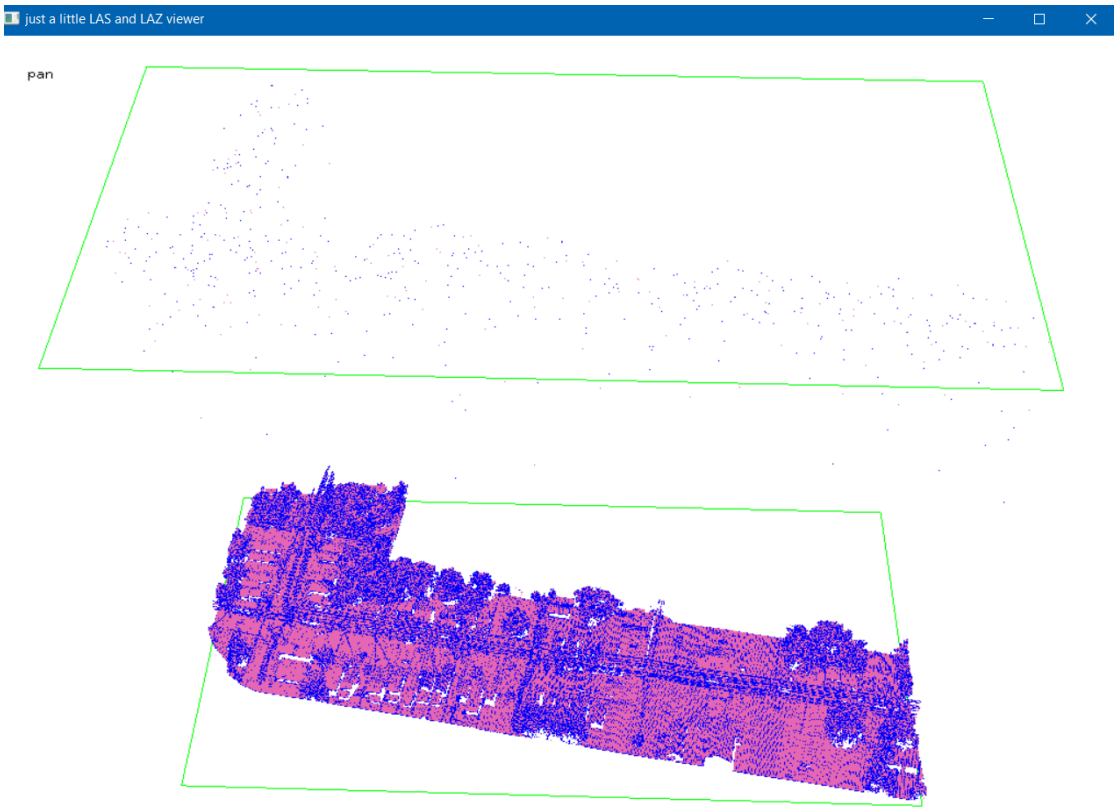
(ii) 10


just a little LAS and LAZ viewer
pan

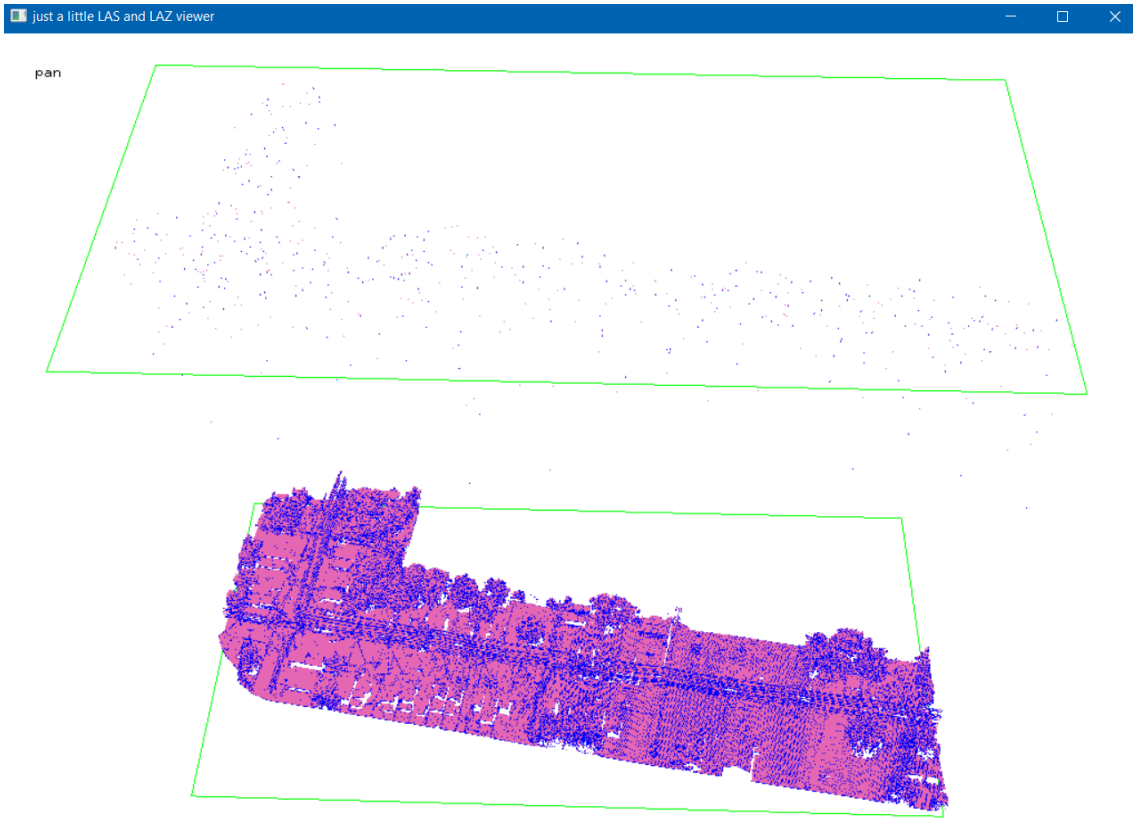
(iii) 50


just a little LAS and LAZ viewer
pan

(iv) Drop line

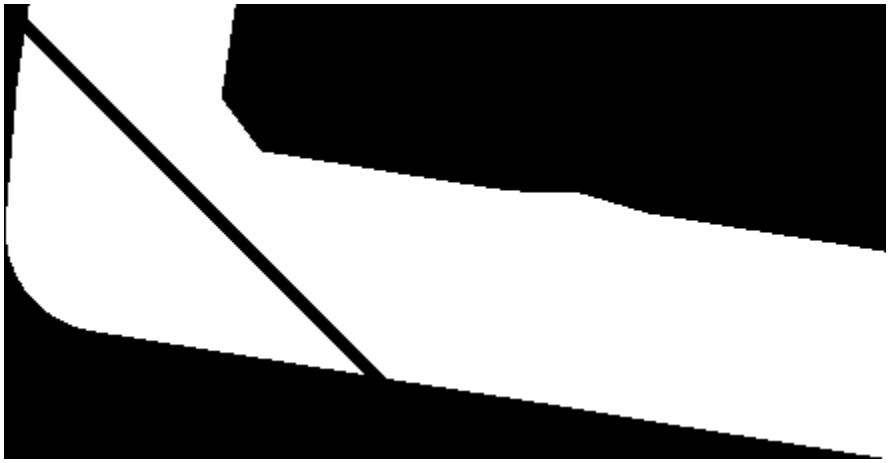


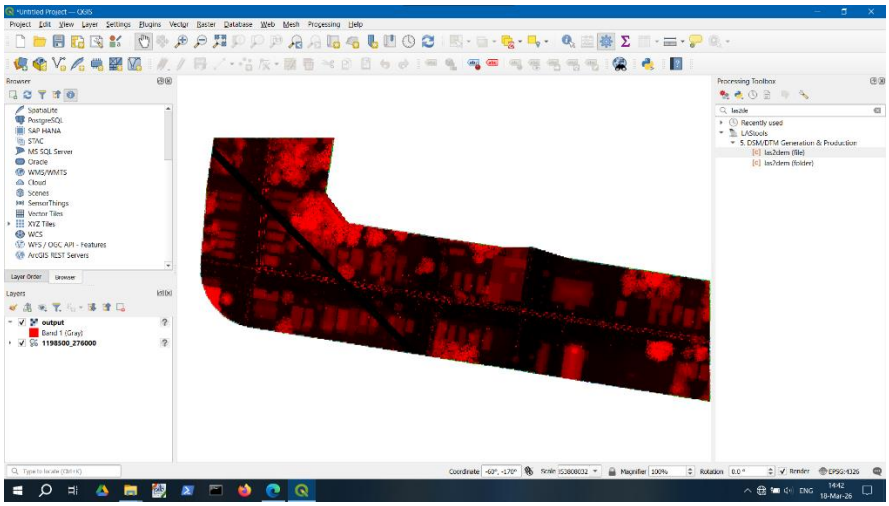
(v) Rise line



• Search for "las2dem" under "file-raster derivatives".
• Brows and Input the file path in "Input LAS/LAZ box" and scroll down to "Save Output" Box Click on button use "Save to File" option to save file as .tif file in default setting. The generated .tif file is the DSM generated from the data at the default resolution of 1.

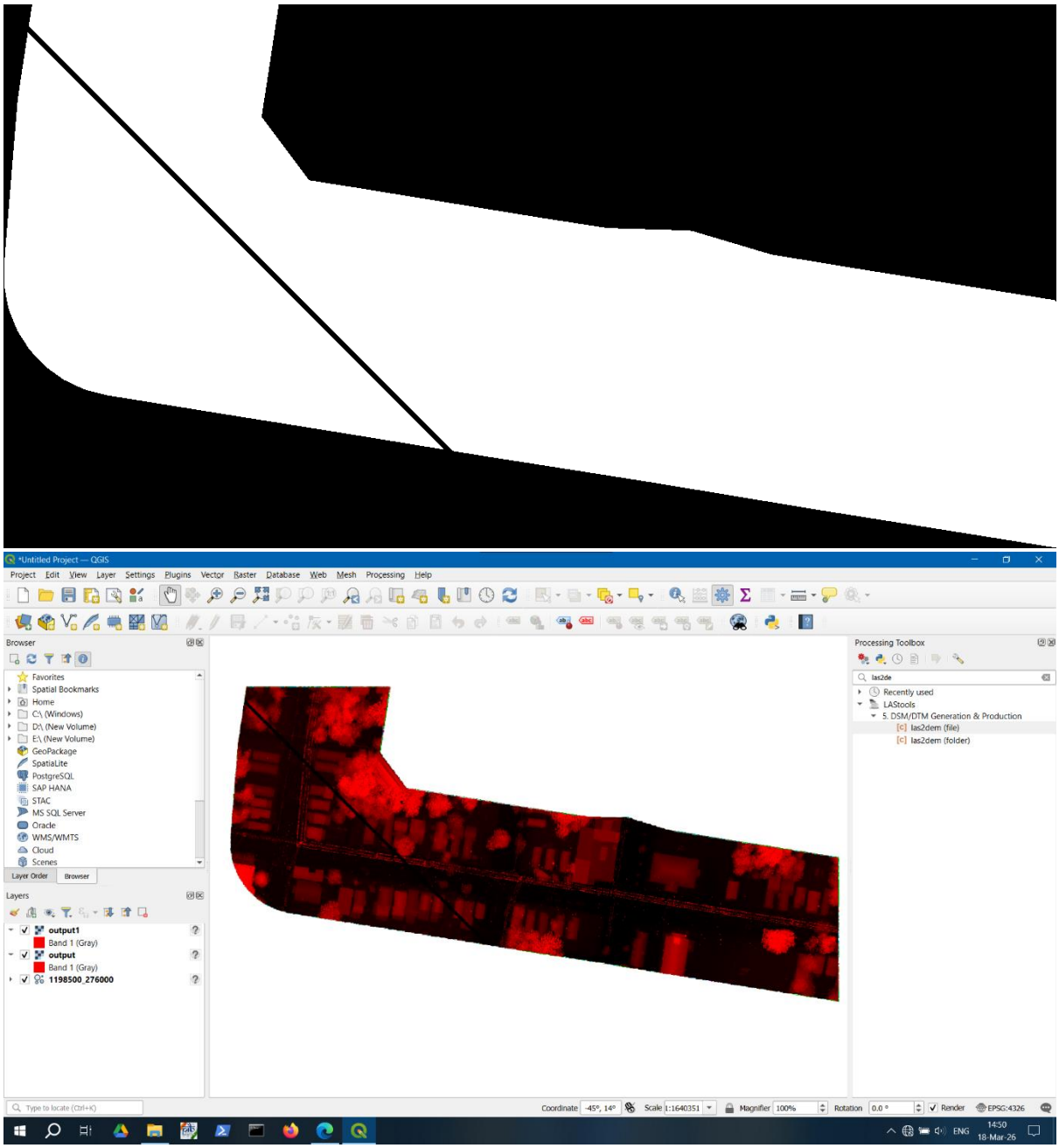
(i) 3



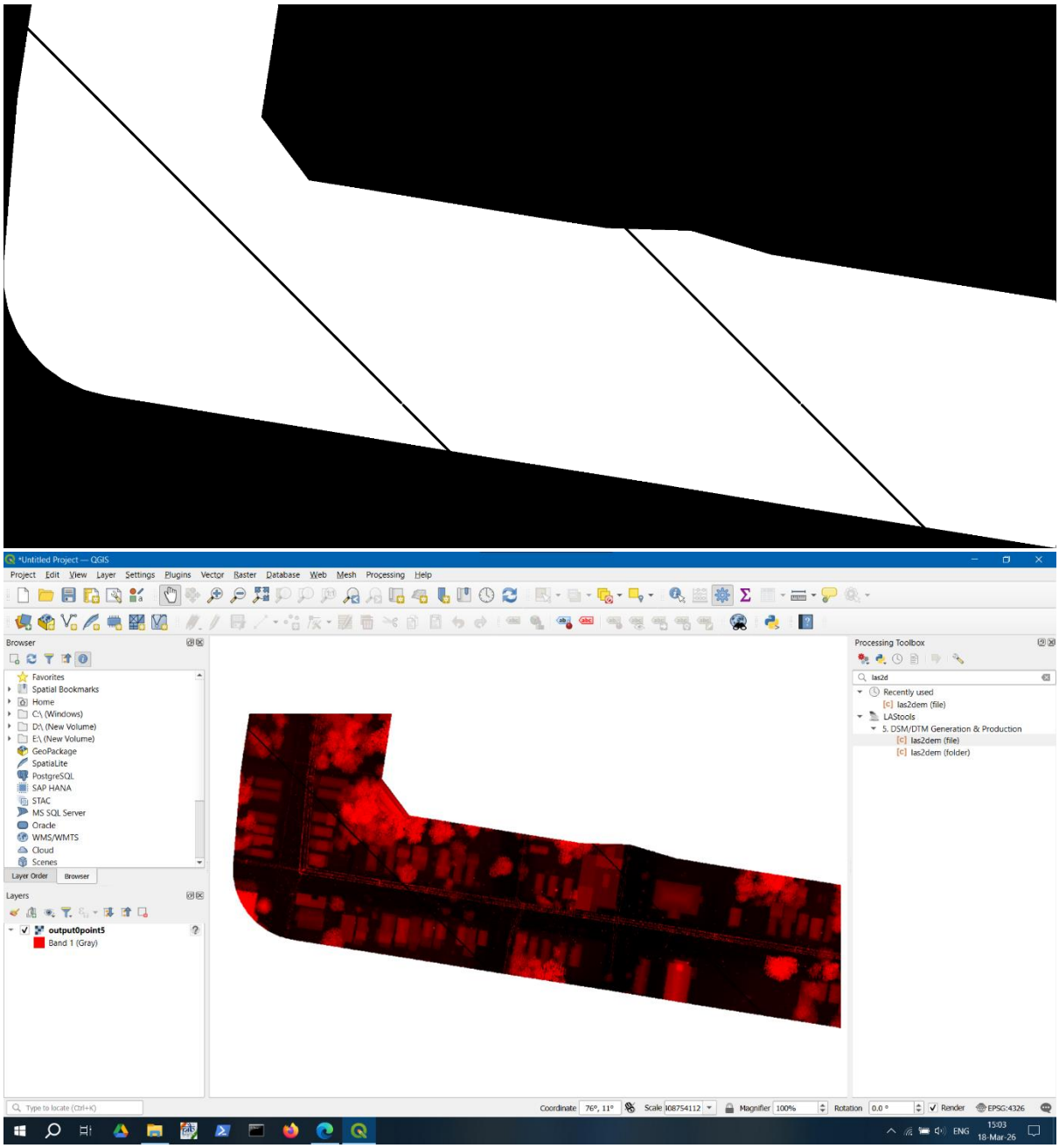


• Change the resolution or pixel size to 0.1 and try to generate a different DSM. You will detect a change in size of the file generated.
• Load the stored .tif files in QGIS using the following steps:
• Select the option open data source manager from the data source manager toolbar located at top left corner.
• Select Raster Tab.
• Give path to the saved .tif file and click add.
• Visualize the different .tif files generated due to varied parameters and report the differences observed and attach the images in the report.

(ii) 1


*Untitled Project — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
Spatial Bookmarks
Home
C:\ (Windows)
D:\ (New Volume)
E:\ (New Volume)
GeoPackage
SpatiaLite
PostgreSQL
SAP HANA
STAC
MS SQL Server
Oracle
WMS/WMTS
Cloud
Scenes
Layer Order
Browser
Layers
output1
Band 1 (Gray)
output
Band 1 (Gray)
1198500_276000
Processing Toolbox
las2de
Recently used
LAStools
5. DSM/DTM Generation & Production
las2dem (file)
las2dem (folder)
Type to locate (Ctrl+K)
Coordinate -45°, 14°
Scale 1:1640351
Magnifier 100%
Rotation 0.0 °
Render
EPSG:4326
ENG
14:50
18-Mar-26

(iii) .5



1. Effect of Changing Resolution on File Size

The lab instructs to change the pixel size to 0.1 to generate a different DSM.

- **Observation:** The file size of the generated .tif file will significantly **increase**.

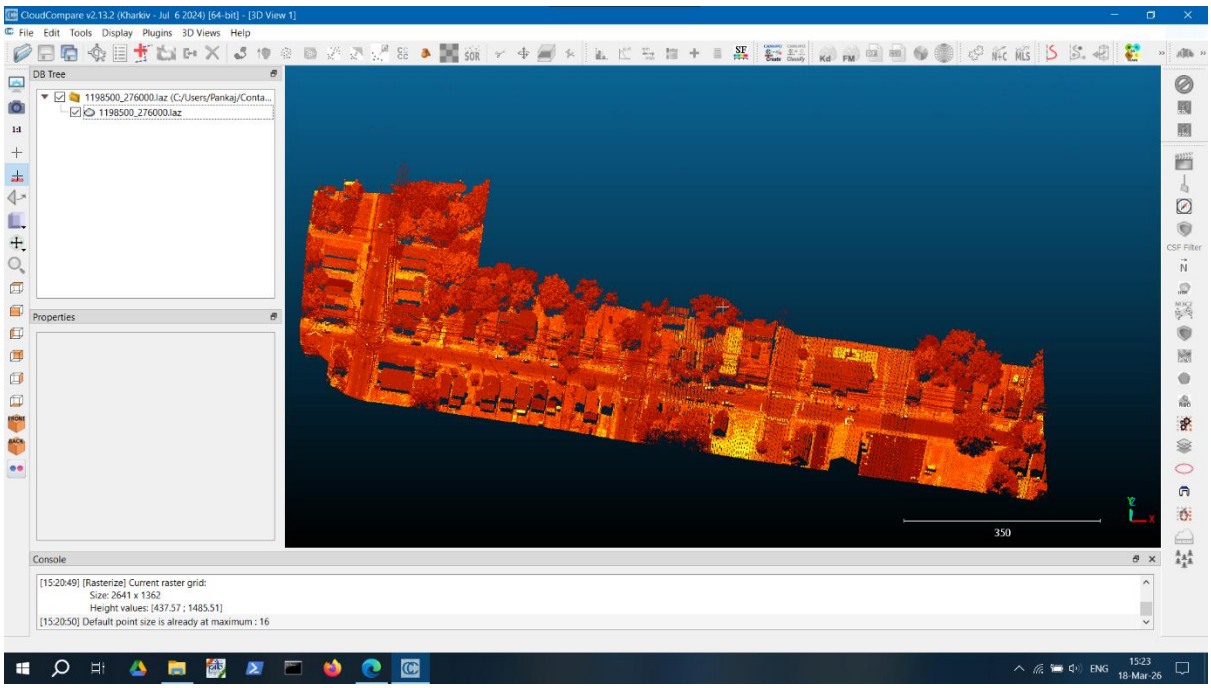
- **Reasoning:** Resolution (or pixel size) dictates the ground area represented by a single pixel. Changing the resolution from default (usually 1 meter) to 0.1 meters means you are reducing the pixel dimensions by a factor of 10. Because area is two-dimensional (x times y), this creates 10 times 10 = 100 times more pixels to cover the exact same geographic area. More pixels require more data storage, directly resulting in a larger .tif file.

• Search for option for generating DSM in cloud compare also and write the procedure in the report and attach the DSM generated from cloud compare also.

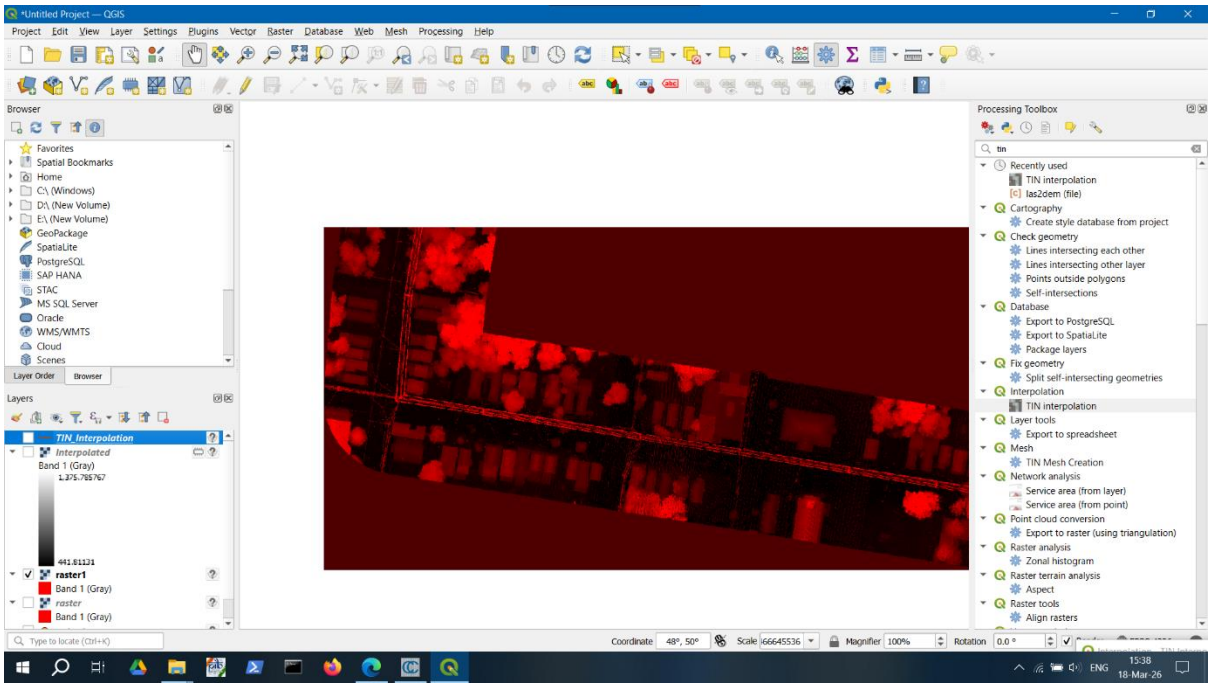
2. DSM Generation Procedure in CloudCompare

The procedure for generating a Digital Surface Model (DSM) using CloudCompare. We can use the following steps:

1. Open CloudCompare and load your .las or .laz point cloud file.
2. Select the loaded point cloud in the "DB Tree" (the left-hand panel).
3. Navigate to the top menu and click on **Tools** > **Projection** > **Rasterize** (or the 2.5D Quadric tool).
4. In the Rasterize tool window, set the **grid step** to your desired resolution (e.g., 1.0 or 0.1).
5. Set the **active layer** to "Height grid values" and ensure the projection direction is set to Z.
6. To create a DSM, set the cell value to **Maximum height** (this captures the top of surfaces like trees and buildings).
7. Click **Update grid** to preview the raster.
8. Click **Export height** or **Raster** to save the generated DSM as a .tif file.



DSM generated by cloud compare opened in Q-GIS is as following



The possible algorithm being used for generating DSM from point cloud.

LAStools (las2dem) and similar software typically use the following approach:

- **Delaunay Triangulation (TIN Generation):** The software first connects the discrete 3D points (X, Y, Z) to form a Triangulated Irregular Network (TIN). This creates a continuous, facet-based 3D surface consisting of non-overlapping triangles.

• Search to "las2tin" under "file-Vector Derivatives " click on it
• Brows and input file path for .las file in INPUT box
• Scroll down to 'Output Box' and save file as .shp
• Load the stored .shp files in QGIS using the following steps:
  • Select the option open data source manager from the data source manager toolbar located at top left corner.
  • Select Vector Tab.
  • Give path to the saved .shp file and click add.
  • Visualize the .shp files generated and report the differences from .tif, observed and attach the images in the report. Observe the .shp file which is basically a TIN (Triangulated Irregular Network). Write a brief note on TIN file and how it is different from simple .las file.

